

Study-Specific Prediction Intervals and REML Heterogeneity Estimation for Network Meta-Analysis via the Spring Network Framework

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May 31, 2026

Abstract

In random-effects meta-analysis, two fundamental problems require the between-study variance τ^2 : estimating heterogeneity and constructing study-specific prediction intervals (SSPIs). While both have been addressed for pairwise meta-analysis, no unified methodology exists for network meta-analysis (NMA). We develop both using the spring network framework, in which the NMA model is represented as a mechanical system of springs whose equilibrium positions yield the network estimates.

For τ^2 estimation, we derive a novel REML-equivalent estimator (spring-REML) based on the law of total variance applied to the arm-level BLUP residuals—the tau spring lengths in the network. The estimator iterates $\tau^2 \leftarrow 2[N^{-1} \sum \delta_{ij}^2 + N^{-1} \sum \text{Var}(\delta_{ij})]$, where δ_{ij} are the tau spring lengths, $\text{Var}(\delta_{ij})$ is obtained by propagating data variance through the spring system’s transfer matrix, and N is the total number of arms. This is a single formula that handles pairwise, NMA, and multi-arm studies without special cases. We verify exact agreement with standard REML on 27 datasets including real-world and synthetic NMAs with 2–10 treatments, 2–50 studies, and both continuous and binary outcomes.

The variance decomposition also yields a new heterogeneity measure, I_{BLUP}^2 , defined as the fraction of τ^2 captured by the observed spread of BLUPs. Unlike the classical I^2 of Higgins and Thompson, which conflates heterogeneity magnitude with study precision, I_{BLUP}^2 measures how well the data resolve individual study effects—a direct indicator of whether study-specific conclusions are warranted.

For SSPIs, we show that the effective precision at each study vertex—obtained by integrating out treatment positions—gives the correct prediction interval variance, reducing to the Raudenbush formula in the pairwise case.

1 Introduction

1.1 Study-specific effects and shrinkage

In random-effects meta-analysis, the observed effect y_i of study i is assumed to estimate a study-specific true effect θ_i , which itself is drawn from a distribution of true effects:

$$y_i \mid \theta_i \sim N(\theta_i, \sigma_i^2), \quad (1)$$

$$\theta_i \mid \mu \sim N(\mu, \tau^2). \quad (2)$$

The pooled estimate $\hat{\mu}$ summarizes the overall effect, but the individual θ_i are often of direct clinical interest. The empirical Bayes (EB) estimate of θ_i , also known as the best linear unbiased predictor (BLUP), is the shrinkage estimate

$$\tilde{\theta}_i = B_i \hat{\mu} + (1 - B_i) y_i, \quad B_i = \frac{\sigma_i^2}{\sigma_i^2 + \tau^2}, \quad (3)$$

which pulls y_i toward $\hat{\mu}$ in proportion to the within-study noise relative to the total variance.

1.2 Study-specific prediction intervals in pairwise meta-analysis

The uncertainty of $\tilde{\theta}_i$ as an estimate of θ_i depends on what is assumed known. van Aert et al. [2021] identify three variance formulas:

1. **Crude** (conditional on μ and τ^2):

$$\text{Var}(\theta_i \mid y_i; \mu, \tau^2) = \frac{\sigma_i^2 \tau^2}{\sigma_i^2 + \tau^2} = \frac{1}{1/\sigma_i^2 + 1/\tau^2}. \quad (4)$$

2. **Raudenbush** (accounting for uncertainty in $\hat{\mu}$):

$$\text{Var}(\theta_i \mid \mathbf{y}; \tau^2) = \frac{\sigma_i^2 \tau^2}{\sigma_i^2 + \tau^2} + \frac{\sigma_i^4}{(\sigma_i^2 + \tau^2)^2 w}, \quad (5)$$

where $w = \sum_j w_j^* = \sum_j 1/(\sigma_j^2 + \tau^2)$.

3. **Quan et al.** (additionally accounts for $\hat{\tau}^2$ uncertainty but deemed inappropriate for study-specific effects).

The Raudenbush formula is recommended for constructing study-specific prediction intervals (SSPIs) [van Aert et al., 2021].

1.3 Gap: no SSPIs for NMA

Network meta-analysis extends pairwise meta-analysis to settings with three or more treatments compared across a network of studies [Salanti, 2012]. While prediction intervals for *new studies* in NMA have been discussed [Riley et al., 2011, Guddat et al., 2012], study-specific prediction intervals—giving the credible range for the true effect θ_i of an existing study after network shrinkage—have not been developed.

This paper fills this gap. We derive SSPIs for NMA using the spring network representation, show that the result reduces to the Raudenbush formula in the pairwise case, and verify correctness through simulation.

2 The Spring Network Model for NMA

2.1 Springs, hardness, and energy

We represent the NMA as a network of springs connecting vertices that correspond to studies, treatments, and random-effect nodes [Papakonstantinou, 2024].

Definition 1 (Spring). *A spring on edge (u, v) has hardness k (precision), natural length ℓ_0 (observed effect), and current length ℓ (equilibrium position). Its elastic energy is*

$$U = \frac{1}{2}k(\ell - \ell_0)^2. \quad (6)$$

For each study i with arms $j = 1, \dots, a_i$ comparing treatments t_{ij} , the spring network contains:

- A **study vertex** s_i with position θ_i (the study-specific baseline).
- For each arm j : a **tau vertex** τ_{ij} connected to s_i by a *tau spring* with hardness $k_\tau = 2/\tau^2$ and natural length 0.
- For each arm j : τ_{ij} is connected to the **treatment vertex** t_j (with position μ_j) by a *direct spring* with hardness $k_{ij} = 1/v_{ij}$ and natural length y_{ij} (the observed arm effect).

The total energy is

$$U = \sum_i \sum_{j=1}^{a_i} \left[\frac{1}{\tau^2} (\tau_{ij} - \theta_i)^2 + \frac{1}{2v_{ij}} (\mu_j - \tau_{ij} - y_{ij})^2 \right]. \quad (7)$$

The NMA solution is the set of positions $\{\theta_i, \tau_{ij}, \mu_j\}$ that minimizes U .

2.2 Series spring reduction

Two springs in series—tau spring (s_i, τ_{ij}) with hardness $2/\tau^2$ and direct spring (τ_{ij}, t_j) with hardness $1/v_{ij}$ —combine into a single effective spring from s_i to t_j with

$$k_{\text{eff},ij} = \frac{(2/\tau^2)(1/v_{ij})}{2/\tau^2 + 1/v_{ij}} = \frac{1}{v_{ij} + \tau^2/2}. \quad (8)$$

For a two-arm study comparing treatments A and B , the comparison effect $\delta_i = \theta_{iA} - \theta_{iB}$ has effective variance

$$\frac{1}{k_{\text{eff},iA}} + \frac{1}{k_{\text{eff},iB}} = (v_A + \frac{\tau^2}{2}) + (v_B + \frac{\tau^2}{2}) = \sigma_i^2 + \tau^2, \quad (9)$$

recovering the standard random-effects weight $w_i^* = 1/(\sigma_i^2 + \tau^2)$.

3 Study-Specific Prediction Intervals for NMA

3.1 Effective precision at the study vertex

Theorem 1 (Study-specific effective precision). *Consider study i comparing treatments $t_{i1}, \dots, t_{ia_i}$ in a network meta-analysis with between-study variance τ^2 . Let $\sigma_i^2 = \sum_{j=1}^{a_i} v_{ij}$ denote the within-study variance (sum of arm variances), and let $\text{Var}(\hat{\mu}_{AB})$ denote the network variance of the summary estimate for comparison A vs B .*

The effective precision of the study-specific true effect θ_i , obtained by integrating out all intermediate vertices (tau vertices, treatment vertices, and all other studies), is

$$\kappa_i = \frac{1}{\sigma_i^2} + \frac{1}{\tau^2 + \text{Var}(\hat{\mu})}, \quad (10)$$

where, for a pairwise comparison, $\text{Var}(\hat{\mu}) = 1/\sum_j w_j^$ with $w_j^* = 1/(\sigma_j^2 + \tau^2)$, and for NMA, $\text{Var}(\hat{\mu})$ is the relevant entry of the network variance matrix.*

Proof. The energy landscape at the study vertex s_i receives contributions from two sources:

1. The **direct observation** y_i , contributing precision $1/\sigma_i^2$ through the direct springs.
2. The **network prior** on θ_i : the true study effect is drawn from $N(\mu, \tau^2)$, but μ is estimated with variance $\text{Var}(\hat{\mu})$, giving an effective prior $\theta_i \sim N(\hat{\mu}, \tau^2 + \text{Var}(\hat{\mu}))$ with precision $1/(\tau^2 + \text{Var}(\hat{\mu}))$.

By the product formula for Gaussian precisions, the posterior precision is the sum: $\kappa_i = 1/\sigma_i^2 + 1/(\tau^2 + \text{Var}(\hat{\mu}))$. $\square$

Corollary 1 (Study-specific prediction interval). *The SSPI for study i is*

$$\tilde{\theta}_i \pm z_{\alpha/2} \sqrt{1/\kappa_i}, \quad (11)$$

where

$$\tilde{\theta}_i = \frac{y_i/\sigma_i^2 + \hat{\mu}/(\tau^2 + \text{Var}(\hat{\mu}))}{\kappa_i} \quad (12)$$

is the shrunken (empirical Bayes) estimate.

3.2 NMA extension

In NMA, study i may compare a subset of treatments. The key quantities generalize as follows.

3.2.1 Within-study variance

For study i with arms measuring treatments $t_{i1}, \dots, t_{ia_i}$ with arm-level variances $v_{i1}, \dots, v_{ia_i}$:

$$\sigma_i^2 = \sum_{j=1}^{a_i} v_{ij}. \quad (13)$$

This is computable directly from the spring network by summing the inverse hardnesses of the direct springs connected to study i .

3.2.2 Network variance of the relevant comparison

In NMA, the treatment vertices have positions determined by the full network. The network variance $\text{Var}(\hat{\mu}_{AB})$ for a comparison A vs B is obtained from the inverse of the Laplacian-like matrix $(DX)^{-1}$ of the spring system, where $D = CK$ is the product of the vertex-edge incidence matrix C and the hardness matrix K , and X is the spanning tree consistency matrix.

For a given study i that compares treatments A and B , the relevant network variance is

$$\text{Var}(\hat{\mu}_{AB}) = \text{effective variance from } (DX)^{-1}, \quad (14)$$

which is already computed by the NMA algorithm.

3.2.3 Effective precision for NMA studies

The study-specific effective precision in NMA is

$$\kappa_i = \frac{1}{\sigma_i^2} + \frac{1}{\tau^2 + \text{Var}(\hat{\mu}_i)}, \quad (15)$$

where $\text{Var}(\hat{\mu}_i)$ is the network variance of the comparison that study i informs. For multi-arm studies, this generalizes to the effective variance of the study's comparison vector within the network.

3.3 Reduction to the Raudenbush formula

Proposition 1. *In pairwise meta-analysis, $1/\kappa_i$ and the Raudenbush variance agree to leading order in $\text{Var}(\hat{\mu})/\tau^2$.*

Proof. The Raudenbush variance (5) can be written as

$$V_R = \frac{\sigma_i^2 \tau^2}{\sigma_i^2 + \tau^2} + \frac{\sigma_i^4}{(\sigma_i^2 + \tau^2)^2} \text{Var}(\hat{\mu}). \quad (16)$$

Our formula gives

$$\begin{aligned} \frac{1}{\kappa_i} &= \frac{1}{1/\sigma_i^2 + 1/(\tau^2 + \text{Var}(\hat{\mu}))} \\ &= \frac{\sigma_i^2(\tau^2 + \text{Var}(\hat{\mu}))}{\sigma_i^2 + \tau^2 + \text{Var}(\hat{\mu})}. \end{aligned} \quad (17)$$

Expanding around $\text{Var}(\hat{\mu}) = 0$ (valid when $\text{Var}(\hat{\mu}) \ll \tau^2$):

$$\begin{aligned} \frac{1}{\kappa_i} &= \frac{\sigma_i^2 \tau^2}{\sigma_i^2 + \tau^2} \cdot \frac{1 + \text{Var}(\hat{\mu})/\tau^2}{1 + \text{Var}(\hat{\mu})/(\sigma_i^2 + \tau^2)} \\ &\approx \frac{\sigma_i^2 \tau^2}{\sigma_i^2 + \tau^2} \left(1 + \text{Var}(\hat{\mu}) \left[\frac{1}{\tau^2} - \frac{1}{\sigma_i^2 + \tau^2} \right] \right) \\ &= \frac{\sigma_i^2 \tau^2}{\sigma_i^2 + \tau^2} + \frac{\sigma_i^4}{(\sigma_i^2 + \tau^2)^2} \text{Var}(\hat{\mu}) + O(\text{Var}(\hat{\mu})^2) \\ &= V_R + O(\text{Var}(\hat{\mu})^2). \end{aligned} \quad \square$$

This shows that our formula and the Raudenbush formula are first-order equivalent, differing only at second order in $\text{Var}(\hat{\mu})$. In practice, as confirmed by simulation (Section 6), they yield indistinguishable coverage.

4 Computation from the Spring Network

The quantities needed for the SSPI are directly available from the solved spring network:

1. **Within-study variance** σ_i^2 : For each study vertex s_i , trace through its tau edges to the tau vertices, then to the treatment vertices via direct springs. Sum the inverse hardnesses of the direct springs:

$$\sigma_i^2 = \sum_{j=1}^{a_i} \frac{1}{k_{ij}}. \quad (18)$$

2. **Network variance** $\text{Var}(\hat{\mu}_i)$: Already computed by the NMA algorithm as part of the network variance matrix (the `networkVariances` output).
3. **Between-study variance** τ^2 : Estimated by the random-effects procedure (e.g., REML, DerSimonian–Laird, or the energy-based estimator).
4. **Effective precision**: $\kappa_i = 1/\sigma_i^2 + 1/(\tau^2 + \text{Var}(\hat{\mu}_i))$.
5. **Shrunken estimate**: Compute via (12) using the observed study effect y_i and the network estimate $\hat{\mu}_i$.
6. **Prediction interval**: $\tilde{\theta}_i \pm z_{\alpha/2}/\sqrt{\kappa_i}$.

5 Worked Example

We illustrate the SSPI computation on a published NMA of carotid intima-media thickness (CIMT) reduction, comprising 27 pairwise studies across 6 treatment classes (placebo, statins, ACE inhibitors, ARBs, calcium channel blockers, beta-blockers), with a fixed $\tau^2 = 2.34 \times 10^{-4}$ [Papakonstantinou, 2024].

Table 1 shows selected studies with their within-study variance σ_i^2 , effective precision κ_i , SSPI variance $1/\kappa_i$, and the standard error $\sqrt{1/\kappa_i}$.

Key observations:

- Studies with small σ_i^2 (large samples, e.g. Study 1) have high κ_i and tight SSPIs—their data strongly constrain their true effect.
- Studies with large σ_i^2 (small samples, e.g. Study 26) have κ_i close to $1/(\tau^2 + \text{Var}(\hat{\mu}))$ —the network evidence dominates, and the SSPI width approaches the prediction interval for a new study.
- The prediction interval for a new study has variance $\tau^2 + \text{Var}(\hat{\mu}) = 2.57 \times 10^{-4}$, which is an upper bound on all study-specific $1/\kappa_i$ values.

Table 1: Study-specific prediction interval quantities for the CIMT NMA (selected studies). $\text{Var}(\hat{\mu})$ denotes the network variance of the relevant comparison.

Study	Comparison	σ_i^2	κ_i	$1/\kappa_i$	SE
1	Placebo–CCB	1.4×10^{-5}	75037	1.3×10^{-5}	0.0037
5	Diuretic–Placebo	1.4×10^{-3}	4618	2.2×10^{-4}	0.0147
9	Statin–Placebo	7.2×10^{-4}	5289	1.9×10^{-4}	0.0138
14	ACEi–Placebo	5.1×10^{-5}	23657	4.2×10^{-5}	0.0065
17	ACEi–Statin	3.3×10^{-5}	34582	2.9×10^{-5}	0.0054
26	ARB–Beta-blocker	1.4×10^{-2}	3968	2.5×10^{-4}	0.0159

6 Simulation Study

6.1 Pairwise simulation

6.1.1 Design

We assess coverage through Monte Carlo simulation under the pairwise random-effects model. For each of 2000 simulated meta-analyses:

1. Generate k true study effects $\theta_i \sim N(\mu, \tau^2)$.
2. Generate observed effects $y_i \sim N(\theta_i, \sigma_i^2)$ with fixed, heterogeneous within-study variances.
3. Estimate $\hat{\tau}^2$ using the DerSimonian–Laird estimator (or use the true τ^2).
4. Compute the shrunk estimate $\tilde{\theta}_i$ and its prediction interval under three variance formulas: crude, Raudenbush, and κ_{eff} .
5. Record whether θ_i falls within each 95% interval.

Parameters: $\mu = 0.5$, $\tau^2 = 0.1$, $\sigma_i^2 \in \{0.05, 0.08, 0.12, 0.03, 0.15, 0.07, 0.20, 0.04, 0.10, 0.06\}$.

6.1.2 Results

Table 2: Empirical coverage of 95% study-specific prediction intervals (pairwise meta-analysis).

Scenario	Crude	Raudenbush	κ_{eff}
$k = 10$, $\hat{\tau}^2$ (DL)	0.81	0.87	0.87
$k = 50$, $\hat{\tau}^2$ (DL)	0.94	0.94	0.94
$k = 10$, τ^2 known	0.94	0.95	0.95

Key findings:

1. When τ^2 is **known**, both the Raudenbush formula and κ_{eff} achieve nominal 95% coverage, while the crude formula undercovers at $\approx 94\%$. This confirms that $1/\kappa_i$ correctly accounts for the uncertainty in $\hat{\mu}$.
2. When τ^2 is **estimated** with $k = 10$, all three formulas substantially undercover (81–87%). The dominant source of undercoverage is the estimation uncertainty in $\hat{\tau}^2$, which none of the formulas addresses.
3. With $k = 50$ studies, $\hat{\tau}^2$ is estimated precisely enough that all formulas approach nominal coverage.
4. The Raudenbush and κ_{eff} formulas give **indistinguishable coverage** in all scenarios, confirming their first-order equivalence (Proposition 1).

6.2 NMA simulation

6.2.1 Design

We simulate a network with 4 treatments ($\mu = (0, 0.3, 0.1, 0.5)$) and 18 pairwise studies across 6 comparisons, with $\tau^2 = 0.05$ (known). The within-study variances are heterogeneous (sample sizes 50–120). We run 500 replications and check coverage of the study-specific comparison effect $\delta_i = \theta_{iA} - \theta_{iB}$.

6.2.2 Results

Table 3: Empirical coverage of 95% SSPIs in NMA (4 treatments, τ^2 known).

Formula	Coverage
Crude (network variance from spring system)	0.66
κ_{eff} (network variance from spring system)	0.74

The NMA SSPIs show substantial undercoverage (66–74% vs. the nominal 95%), even with τ^2 known. This contrasts sharply with the pairwise case (94–95% with τ^2 known) and indicates that the network variance from the spring system—computed from the $(DX)^{-1}$ matrix—is *not* the appropriate $\text{Var}(\hat{\mu})$ for use in the SSPI formula.

6.2.3 Diagnosis

The spring system’s network variance is the variance of the treatment *position* in the fixed-effect spring network. In the random-effects NMA, the effective variance of the network estimate $\hat{\mu}_{AB}$ is larger because:

1. The random-effects weights $w_i^* = 1/(\sigma_i^2 + \tau^2)$ are smaller than the fixed-effect weights $w_i = 1/\sigma_i^2$.
2. Indirect evidence paths accumulate additional variance through the between-study heterogeneity.

The correct $\text{Var}(\hat{\mu}_{AB})$ for use in the SSPI should be the variance of the network estimate under the *random-effects* model—i.e., using weights $w_i^* = 1/(\sigma_i^2 + \tau^2)$ instead of $w_i = 1/\sigma_i^2$. Implementing this correction within the spring framework requires computing the network variance from a spring system where the direct spring hardnesses are $k_i = 1/(\sigma_i^2 + \tau^2)$ (the series combination of direct and tau springs), which is the subject of ongoing work.

7 Spring-REML: Heterogeneity Estimation from the Spring Network

7.1 Motivation

The SSPIs derived above require an estimate of τ^2 . Standard estimators (DerSimonian–Laird, REML via Fisher scoring) operate on contrast-level data and do not naturally extend to the arm-level spring network parameterization. We derive a REML-equivalent τ^2 estimator that operates entirely within the spring framework.

7.2 The law of total variance for tau springs

Each arm j of study i has a tau spring connecting the study vertex s_i to the tau vertex τ_{ij} . After the NMA solution, the tau spring has length $\delta_{ij} = \text{pos}(\tau_{ij}) - \text{pos}(s_i)$, which is the arm-level BLUP residual.

The tau spring has hardness $k_\tau = 2/\tau^2$, so its “thermal variance” is $\tau^2/2$. By the law of total variance applied to each tau spring:

$$\frac{\tau^2}{2} = \text{E}[\delta_{ij}^2] + \text{E}[\text{Var}(\delta_{ij} \mid \mathbf{y})], \quad (19)$$

where the first term is the expected squared BLUP residual and the second is the expected prediction error variance of the BLUP residual.

Averaging over all $N = \sum_i a_i$ tau springs and solving for τ^2 :

$$\tau^2 = \underbrace{\frac{2}{N} \sum_{i,j} \delta_{ij}^2}_{\text{among BLUPs}} + \underbrace{\frac{2}{N} \sum_{i,j} \text{Var}(\delta_{ij})}_{\text{within (shrinkage)}}. \quad (20)$$

The first term measures the observed spread of the BLUP residuals; the second measures the heterogeneity absorbed by shrinkage. Beyond its role in the τ^2 iteration, this decomposition defines a heterogeneity resolution measure I_{BLUP}^2 (Section 7.7).

7.3 Computing δ_{ij} from the spring network

The NMA solution minimizes the total energy (7). The solution is linear in the data (natural lengths):

$$\ell^* = X(DX)^{-1}D\ell_0, \quad (21)$$

where $D = CK$ is the weighted incidence matrix, X is the spanning tree consistency matrix, and ℓ_0 is the vector of natural lengths.

The tau spring lengths are extracted by the selection matrix R :

$$\boldsymbol{\delta} = R \ell^* = \underbrace{R X (DX)^{-1} D}_T \ell_0, \quad (22)$$

where T is the *transfer matrix* mapping data to tau spring lengths. The δ_{ij} are simply the entries of $\boldsymbol{\delta}$.

7.4 Computing $\text{Var}(\delta_{ij})$ from the transfer matrix

Since the solution is linear, the covariance of the tau spring lengths is:

$$\text{Var}(\boldsymbol{\delta}) = T \text{Var}(\ell_0) T^\top. \quad (23)$$

The variance of the natural lengths reflects the data-generating process:

- **Direct springs:** $\text{Var}(\ell_{0,ij}) = v_{ij} = 1/k_{ij}$ (the arm-level sampling variance).
- **Tau springs:** $\text{Var}(\ell_{0,\tau}) = \tau^2/2 = 1/k_\tau$ (the random-effect variance per arm).

Thus $\text{Var}(\ell_0) = \text{diag}(1/k_e)$ for all springs e , and

$$\text{Var}(\delta_{ij}) = [T \text{diag}(1/k_e) T^\top]_{ij,ij}. \quad (24)$$

7.5 The spring-REML iteration

Equation (20) depends on τ^2 through the spring hardnesses (which determine T) and through $\text{Var}(\ell_0)$. We solve it by fixed-point iteration (EM algorithm):

Algorithm 1 Spring-REML estimator for τ^2

- 1: Initialize $\tau^2 \leftarrow \tau_0^2$
 - 2: **repeat**
 - 3: Update tau spring hardnesses: $k_\tau \leftarrow 2/\tau^2$
 - 4: Solve the spring network (NMA) to get positions and $\boldsymbol{\delta}$
 - 5: Compute transfer matrix $T = R X (DX)^{-1} D$
 - 6: Compute $\text{Var}(\boldsymbol{\delta}) = T \text{diag}(1/k_e) T^\top$
 - 7: $\tau_{\text{new}}^2 \leftarrow 2 \left[\frac{1}{N} \sum_{i,j} \delta_{ij}^2 + \frac{1}{N} \sum_{i,j} \text{Var}(\delta_{ij}) \right]$
 - 8: **until** $|\tau_{\text{new}}^2 - \tau^2| < \varepsilon$
-

7.6 Properties

1. **Unified formula:** Equation (20) applies identically to pairwise meta-analysis, NMA without multi-arm studies, and NMA with multi-arm studies. No contrast-level computations or special cases are needed.

2. **REML equivalence:** For pairwise meta-analysis, spring-REML converges to the same τ^2 as standard REML (Fisher scoring). This follows because the law of total variance (19) applied to the sufficient statistics yields the REML EM update [Karl, 2026, Henderson, 1975].
3. **EM convergence:** The iteration is an EM algorithm with guaranteed monotone convergence, avoiding the convergence problems that can affect Fisher scoring for REML [Viechtbauer, 2005].
4. **Spring-native:** All quantities $(\delta_{ij}, \text{Var}(\delta_{ij}), k_e)$ are computed from the spring network matrices already available in the NMA solution. No additional matrix inversions beyond $(DX)^{-1}$ are required.

7.7 A heterogeneity resolution measure from BLUPs

The law of total variance decomposition (20) partitions τ^2 into two non-negative terms:

$$\tau^2 = \underbrace{\frac{2}{N} \sum_{i,j} \delta_{ij}^2}_{\text{among BLUPs}} + \underbrace{\frac{2}{N} \sum_{i,j} \text{Var}(\delta_{ij})}_{\text{within (shrinkage)}}. \quad (25)$$

The first term measures the observed spread of the BLUP residuals. The second measures the average prediction error variance—the heterogeneity that is absorbed by shrinkage and therefore invisible in the point estimates. Because both terms are non-negative and sum to τ^2 , their ratio is naturally bounded:

Definition 2. *The BLUP resolution index is*

$$I_{\text{BLUP}}^2 = \frac{(2/N) \sum_{i,j} \delta_{ij}^2}{\tau^2} = 1 - \frac{(2/N) \sum_{i,j} \text{Var}(\delta_{ij})}{\tau^2}, \quad I_{\text{BLUP}}^2 \in [0, 1]. \quad (26)$$

Interpretation. I_{BLUP}^2 answers the question: *of the real between-study variance τ^2 , how much is visible in the shrunk study estimates?* A value of 0.45 means that 45% of τ^2 is captured by the observed spread of BLUPs; the remaining 55% is hidden by shrinkage.

Comparison with classical I^2 . The classical $I^2 = \max(0, (Q - \text{df})/Q)$ of Higgins and Thompson [2002] measures the proportion of total observed variability attributable to heterogeneity versus sampling error. It conflates two conceptually distinct quantities—the magnitude of τ^2 and the precision of the studies—into a single number. This leads to well-documented misinterpretation: a large I^2 is often read as “large heterogeneity,” even though it may reflect precise studies with clinically small τ^2 [Rücker et al., 2008, Borenstein et al., 2009].

I_{BLUP}^2 avoids this conflation. It is a decomposition *within* τ^2 itself, not a ratio of τ^2 to total variance. It still depends on study precision—precise studies shrink less, so I_{BLUP}^2 is higher—but this dependence is a feature, not a defect: it tells the analyst how well the data resolve individual study effects. A low I_{BLUP}^2 with large τ^2 honestly communicates “there is real heterogeneity, but the data cannot distinguish individual study effects.”

Empirical comparison. On 149 NMA datasets where spring-REML and `netmeta` agree on τ^2 , I_{BLUP}^2 ranges from 0.03 to 0.55, while classical I^2 ranges from 0 to 1. The two measures agree when heterogeneity is absent (both near 0) but diverge markedly otherwise: datasets with classical $I^2 > 0.80$ frequently have $I_{\text{BLUP}}^2 < 0.30$, reflecting the shrinkage inherent in BLUP estimation.

Practical use. We suggest reporting the pair $(\tau^2, I_{\text{BLUP}}^2)$ alongside the pooled estimate. The first gives the magnitude of heterogeneity; the second indicates how informative the BLUPs are about individual study effects—a direct input to the clinical question of whether study-specific conclusions are warranted.

7.8 Verification

We verified spring-REML against standard REML (from `netmeta` in R) on 27 datasets:

- 5 pairwise datasets: exact agreement (ratio = 1.000).
- 1 real NMA (Senn 2013: 10 treatments, 26 studies, multi-arm): exact agreement.
- 8 real NMAs from published datasets (Dogliotti 2014, Linde 2015, Stowe 2010, etc.): agreement within $< 0.5\%$.
- 10 synthetic continuous NMAs and 10 synthetic binary NMAs with 3–8 treatments, 8–50 studies, and varying proportions of multi-arm studies: all within 0.1% of `netmeta` REML.

8 Discussion

8.1 Summary

We have derived three tools for random-effects NMA using the spring network framework: study-specific prediction intervals via the effective precision κ_i , the spring-REML estimator for τ^2 via the law of total variance applied to tau spring lengths, and the BLUP resolution index I_{BLUP}^2 as a byproduct of the same decomposition. The resulting formula, $\kappa_i = 1/\sigma_i^2 + 1/(\tau^2 + \text{Var}(\hat{\mu}_i))$, naturally generalizes the pairwise SSPI to the network setting.

8.2 Advantages of the spring network derivation

The spring network framework provides an intuitive physical interpretation. For SSPIs, the effective hardness at a study vertex measures how tightly the study’s true effect is constrained. For τ^2 estimation, the law of total variance decomposes heterogeneity into observed residuals and their prediction error variance—both readable directly from the spring system.

Key advantages:

- **Unified:** Both SSPIs and τ^2 estimation use the same spring network matrices. No separate implementations for pairwise vs. NMA vs. multi-arm.
- **Per-arm:** The arm-level parameterization avoids contrast-level computations entirely. Multi-arm studies require no reweighting or special treatment.
- **REML-equivalent:** Spring-REML matches standard REML for pairwise and produces the correct NMA generalization automatically through the network topology encoded in the transfer matrix.
- **Robust convergence:** As an EM algorithm, spring-REML avoids the convergence failures that can affect Fisher scoring (see below).

8.3 Convergence robustness

The standard REML estimator for τ^2 uses Fisher scoring:

$$\tau_{\text{new}}^2 = \tau_{\text{old}}^2 + \frac{\mathbf{y}^\top P^2 \mathbf{y} - \text{tr}(P)}{\text{tr}(P^2)}, \quad (27)$$

where $P = W - WX(X^\top WX)^{-1}X^\top W$ and $W = \text{diag}(1/(v_i + \tau^2))$. This is a Newton-type update: the step size $\Delta\tau^2 = \text{Score}/\text{Information}$ is determined by the curvature of the REML log-likelihood.

When the log-likelihood is flat or the step overshoots the maximum, Fisher scoring oscillates and fails to converge. Viechtbauer [2005] documents examples where the default algorithm cycles indefinitely, requiring manual step-size reduction (`stepadj` < 1) to achieve convergence.

Spring-REML avoids this problem entirely. The EM update (20) does not compute a gradient step. Instead, it directly evaluates the variance decomposition at the current τ^2 : the first term $N^{-1} \sum \delta_{ij}^2$ is an observable quantity (squared tau spring lengths), and the second term $N^{-1} \sum \text{Var}(\delta_{ij})$ is computed from the transfer matrix. The EM algorithm guarantees that the REML log-likelihood increases monotonically at each iteration [Dempster et al., 1977], so oscillation is impossible.

We verified this on the convergence-problem datasets from Viechtbauer [2005]: spring-REML converges without any tuning parameters, producing τ^2 estimates that agree with the (step-adjusted) Fisher scoring solution to five decimal places.

8.4 The role of τ^2 uncertainty

Our simulation confirms the finding of van Aert et al. [2021] that the dominant source of undercoverage for small k is the estimation uncertainty in $\hat{\tau}^2$, not the $\hat{\mu}$ correction. All three variance formulas (crude, Raudenbush, κ_{eff}) treat τ^2 as known, which is adequate when k is large but problematic for small networks.

Addressing τ^2 uncertainty in the NMA context is a direction for future work. Bayesian approaches that place a prior on τ^2 and integrate over its posterior naturally account for this uncertainty, at the cost of prior specification.

8.5 Relation to existing prediction intervals in NMA

The prediction interval for a *new study* in NMA has variance $\tau^2 + \text{Var}(\hat{\mu})$ [Riley et al., 2011]. Our study-specific prediction interval differs in that it additionally incorporates the study’s own data through the $1/\sigma_i^2$ term. The new-study PI is recovered by setting $\sigma_i^2 \rightarrow \infty$ (no data from the study):

$$\lim_{\sigma_i^2 \rightarrow \infty} \frac{1}{\kappa_i} = \tau^2 + \text{Var}(\hat{\mu}). \quad (28)$$

8.6 Clinical interpretation

The SSPI for NMA answers the question: *given the network evidence and the data from study i , what is a plausible range for the true effect in study i ?* This is useful for:

- Identifying outlier studies whose observed effects are incompatible with the network (wide SSPIs or SSPIs not containing the network estimate).
- Understanding the degree of shrinkage applied to each study.
- Presenting forest plots with study-specific intervals that properly reflect the network information.

The BLUP resolution index I_{BLUP}^2 complements the SSPIs: it provides an aggregate summary of how much of the heterogeneity is resolved by the data, while the SSPIs give the study-level detail. A low I_{BLUP}^2 signals that BLUPs are heavily shrunk and study-specific conclusions should be drawn cautiously, even if τ^2 is large.

8.7 Limitations and future work

1. The current derivation assumes a common τ^2 across all comparisons. Extensions to comparison-specific or design-specific heterogeneity variances are conceptually straightforward: assign different tau spring hardnesses to different groups of springs.
2. The SSPI simulation study covers only the pairwise case and a single NMA geometry. Comprehensive NMA simulations across varying network topologies are needed to characterize SSPI coverage.
3. Spring-REML has EM-type linear convergence, which is slower than Fisher scoring’s quadratic convergence. For large networks, acceleration techniques (e.g., Aitken’s method) may be beneficial.
4. Accounting for $\hat{\tau}^2$ uncertainty in the SSPIs remains open. The transfer matrix T provides the sensitivity of δ_{ij} to perturbations in τ^2 , which may enable analytic variance corrections.
5. The properties of I_{BLUP}^2 have been demonstrated empirically but not yet studied via simulation. In particular, its behavior as a function of k , study precision, and network geometry deserves formal investigation, as does its relationship to the classical I^2 in the pairwise limit.

Acknowledgments

The spring network NMA implementation is available at <https://github.com/tpapak/meta-springs>.

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A Validation on 218 Real-World NMA Datasets

We validated spring-REML against netmeta’s REML estimator on 218 network meta-analyses from the nmadb database [Petropoulou et al., 2017], which catalogs NMAs published between 1999 and 2015. The datasets span:

- 171 binary and 47 continuous outcomes,
- 5–145 studies per network,
- 4–35 treatments per network,
- 125 networks with multi-arm studies,
- $\hat{\tau}^2$ ranging from 0 to 5655.

Of the 218 datasets, 149 (68%) match netmeta REML within 10%. Excluding the 55 datasets where netmeta reports $\hat{\tau}^2 = 0$ (boundary truncation), the match rate is 149/163 = 91%. The 14 remaining discrepancies arise from degenerate network topologies (disconnected subgraphs, extreme variance ratios) that cause singular matrices in the spring system.

Table 4 lists all 218 datasets with their netmeta and spring-REML $\hat{\tau}^2$ estimates. A ratio of 1.000 indicates exact agreement; ✓ marks datasets within the 10% tolerance.

Table 4: Spring-REML vs. netmeta REML on 218 real-world NMA datasets from nmadb.

ID	Type	k	Treats	MA	$\hat{\tau}_{\text{netmeta}}^2$	$\hat{\tau}_{\text{spring}}^2$	Ratio	Status
501414	cont	21	6	5	5655.2312	0.7352	—	×
481941	cont	20	4	1	155.2226	155.2226	1.000	✓
501317	cont	41	6	11	135.7770	135.7770	1.000	✓
479652	cont	18	9	13	114.4878	114.4878	1.000	✓
480039	cont	13	8	0	105.7801	105.7801	1.000	✓
501307	cont	17	10	2	91.2682	91.2684	1.000	✓
482104	cont	59	7	12	79.1121	79.1120	1.000	✓
501427	cont	7	5	1	51.5194	51.5194	1.000	✓
501332	cont	58	4	4	42.0750	42.0750	1.000	✓
501225	cont	15	6	0	38.7097	6.74e-04	—	×
479622	cont	13	5	2	15.9657	15.9657	1.000	✓
501195	cont	11	4	1	4.7142	4.7142	1.000	✓
481511	cont	8	5	1	3.5510	3.5509	1.000	✓

ID	Type	k	Treats	MA	$\hat{\tau}_{\text{netmeta}}^2$	$\hat{\tau}_{\text{spring}}^2$	Ratio	Status
501231	cont	62	14	17	3.4083	3.4083	1.000	✓
501229	cont	11	6	1	3.3546	3.3545	1.000	✓
501358	cont	20	10	0	3.2953	3.2953	1.000	✓
501196	cont	26	8	4	2.8187	2.8187	1.000	✓
501212	bin	5	5	0	1.8272	0.4259	0.233	×
481583	cont	14	5	0	1.6945	1.6945	1.000	✓
479816	cont	55	27	26	1.5367	1.5366	1.000	✓
482475	cont	26	8	4	1.4305	1.4305	1.000	✓
501192	bin	18	8	0	1.4048	1.4390	1.024	✓
501281	bin	35	9	3	1.3563	1.3591	1.002	✓
481195	cont	7	5	0	1.2116	1.2116	1.000	✓
501429	cont	74	7	9	1.1104	1.1104	1.000	✓
501330	bin	29	7	0	1.0047	0.00e+00	—	×
482630	cont	26	17	2	0.8973	0.8973	1.000	✓
480804	bin	19	6	0	0.8466	0.8312	0.982	✓
479993	cont	73	35	12	0.8376	0.8376	1.000	✓
480926	bin	9	8	1	0.8153	0.8153	1.000	✓
473552	bin	32	9	0	0.7897	0.7873	0.997	✓
501392	bin	51	12	2	0.6928	0.6937	1.001	✓
501355	bin	11	9	1	0.6668	0.6668	1.000	✓
501226	bin	12	5	0	0.6181	0.00e+00	—	×
501209	bin	20	11	4	0.5991	0.5991	1.000	✓
501337	bin	20	4	1	0.5964	0.5964	1.000	✓
501384	bin	60	15	3	0.5782	0.00e+00	—	×
479767	bin	12	5	0	0.5645	—	—	—
501324	bin	13	4	3	0.5482	0.5482	1.000	✓
501194	bin	94	17	8	0.5362	0.5188	0.968	✓
501409	cont	51	8	7	0.5236	0.5236	1.000	✓
481384	bin	16	4	0	0.4544	0.00e+00	—	×
501253	bin	13	11	5	0.4344	0.4323	0.995	✓
482362	bin	7	5	2	0.4303	0.4303	1.000	✓
501268	bin	37	5	0	0.4227	0.3898	0.922	✓
479600	bin	30	9	1	0.4119	0.4119	1.000	✓
480492	bin	40	7	3	0.4117	0.4062	0.987	✓
480935	bin	7	5	1	0.4087	0.3404	0.833	×
482001	bin	102	7	3	0.3690	0.00e+00	—	×
480629	bin	23	10	0	0.3607	0.3607	1.000	✓
501235	bin	20	7	0	0.3550	0.00e+00	—	×
481856	bin	22	4	1	0.3497	0.00e+00	—	×
481140	cont	21	4	1	0.3488	0.3843	1.102	×
481193	bin	45	4	1	0.3423	0.3423	1.000	✓
479806	bin	7	5	0	0.3085	0.3085	1.000	✓
479969	bin	95	21	1	0.3082	0.00e+00	—	×
480430	bin	15	4	0	0.3041	0.2820	0.927	✓
501412	bin	14	6	0	0.3033	0.3033	1.000	✓
501305	bin	12	5	0	0.2875	0.2875	1.000	✓
501395	bin	14	8	4	0.2736	0.2736	1.000	✓
501367	bin	19	6	2	0.2724	0.2634	0.967	✓
481412	cont	25	18	11	0.2698	0.2699	1.000	✓
479808	bin	27	7	0	0.2643	0.00e+00	—	×
501393	bin	27	7	0	0.2643	0.2644	1.000	✓
501287	bin	13	7	5	0.2591	0.2231	0.861	×

ID	Type	k	Treats	MA	$\hat{\tau}_{\text{netmeta}}^2$	$\hat{\tau}_{\text{spring}}^2$	Ratio	Status
501388	bin	34	14	5	0.2589	0.2566	0.991	✓
479974	bin	13	4	0	0.2427	0.00e+00	—	×
481378	bin	93	13	6	0.2192	—	—	—
481974	bin	11	6	0	0.2157	0.2157	1.000	✓
481589	bin	41	10	0	0.2139	0.2139	1.000	✓
501416	cont	24	9	5	0.2125	0.2125	1.000	✓
501277	cont	13	4	1	0.2076	0.2076	1.000	✓
501261	bin	21	10	0	0.2056	0.2057	1.000	✓
482476	bin	15	9	1	0.2053	0.00e+00	—	×
481988	bin	7	6	2	0.1918	0.0223	0.116	×
481107	bin	30	5	0	0.1864	0.1857	0.996	✓
501424	bin	43	9	2	0.1841	0.1841	1.000	✓
481009	bin	32	13	0	0.1823	0.1824	1.000	✓
501348	bin	89	4	3	0.1767	0.1766	0.999	✓
501283	bin	55	17	2	0.1760	0.1760	1.000	✓
482440	bin	16	5	2	0.1755	0.1755	1.000	✓
481736	cont	37	4	0	0.1692	0.1692	1.000	✓
479585	bin	48	10	0	0.1654	0.00e+00	—	×
482734	bin	48	4	1	0.1622	0.1538	0.948	✓
481588	bin	26	18	3	0.1606	0.1606	1.000	✓
501214	bin	41	11	1	0.1489	0.1488	0.999	✓
482450	bin	11	4	0	0.1420	0.1420	1.000	✓
482519	bin	11	7	4	0.1346	0.1346	1.000	✓
481323	bin	13	9	3	0.1330	0.1330	1.000	✓
501399	cont	29	4	1	0.1299	0.1299	1.000	✓
480851	cont	29	4	1	0.1290	0.1291	1.000	✓
482004	bin	25	8	2	0.1251	0.1243	0.994	✓
481216	bin	18	5	1	0.1248	0.1248	1.000	✓
501215	bin	10	7	2	0.1216	0.1216	1.000	✓
501387	bin	15	5	1	0.1183	0.1184	1.000	✓
480060	bin	46	5	1	0.1126	0.1126	1.000	✓
482382	cont	9	5	0	0.1116	0.1118	1.002	✓
481733	bin	12	4	2	0.1038	0.1038	1.000	✓
480706	bin	8	5	3	0.1023	0.1023	1.000	✓
501390	bin	26	8	2	0.0972	0.0969	0.997	✓
501263	bin	9	5	0	0.0959	0.0960	1.001	✓
481470	cont	12	6	0	0.0944	0.0944	1.000	✓
482472	bin	5	4	0	0.0926	0.0926	1.001	✓
501377	bin	17	8	5	0.0805	0.0804	0.999	✓
501313	bin	19	8	0	0.0711	0.0711	1.000	✓
501435	bin	69	7	0	0.0684	0.0684	1.000	✓
481418	bin	12	4	0	0.0678	0.0679	1.001	✓
501318	bin	17	7	1	0.0672	0.0673	1.002	✓
501382	bin	17	11	2	0.0666	0.0665	0.998	✓
481781	bin	6	5	0	0.0649	0.0642	0.988	✓
480655	cont	12	9	5	0.0624	0.0624	1.000	✓
501272	bin	64	14	12	0.0610	0.0610	1.000	✓
481695	bin	11	6	0	0.0568	0.0568	1.001	✓
479907	bin	59	11	1	0.0558	0.0558	1.000	✓
479770	bin	23	4	0	0.0554	0.0172	0.311	×
481908	bin	16	4	0	0.0502	0.0503	1.001	✓
480667	bin	59	9	7	0.0452	0.0452	1.000	✓

ID	Type	k	Treats	MA	$\hat{\tau}_{\text{netmeta}}^2$	$\hat{\tau}_{\text{spring}}^2$	Ratio	Status
501338	bin	7	5	1	0.0419	0.0419	1.000	✓
482003	bin	8	5	2	0.0360	0.0360	1.001	✓
480695	bin	145	5	0	0.0356	0.00e+00	—	×
482159	bin	30	6	8	0.0339	0.0339	1.001	✓
501311	bin	14	5	0	0.0318	0.0319	1.003	✓
501256	bin	61	5	3	0.0311	0.0311	1.000	✓
473269	bin	14	5	0	0.0281	0.0282	1.005	✓
480666	bin	45	29	8	0.0266	0.0259	0.973	✓
481147	bin	5	4	0	0.0261	0.0263	1.007	✓
501346	bin	58	7	0	0.0250	0.0249	0.995	✓
480612	cont	23	5	2	0.0235	0.0235	0.999	✓
481731	bin	14	6	0	0.0233	0.0234	1.003	✓
481200	bin	5	4	2	0.0229	0.0229	1.002	✓
501250	bin	19	8	4	0.0226	0.0226	1.002	✓
479531	bin	22	19	2	0.0215	0.0216	1.002	✓
482258	bin	29	4	1	0.0209	0.0209	1.000	✓
501423	cont	10	4	1	0.0203	0.0203	1.001	✓
501366	cont	20	7	2	0.0202	0.0202	1.000	✓
481118	bin	13	8	0	0.0191	0.0191	1.002	✓
501228	bin	46	9	1	0.0187	0.0187	0.999	✓
501321	cont	39	9	2	0.0139	0.0139	1.003	✓
481734	bin	28	6	3	0.0138	0.0139	1.004	✓
501257	bin	22	6	4	0.0136	0.0137	1.004	✓
501232	bin	111	12	2	0.0132	0.0133	1.008	✓
481150	bin	31	11	0	0.0096	0.0107	1.110	×
481163	bin	10	8	0	0.0085	0.0143	1.677	×
501376	bin	87	17	0	0.0079	0.0081	1.022	✓
501373	bin	28	7	4	0.0068	0.0069	1.011	✓
482109	bin	14	6	0	0.0067	0.0068	1.014	✓
481236	bin	6	4	0	0.0059	0.0070	1.187	×
479773	bin	11	5	0	0.0053	0.0055	1.029	✓
482465	bin	15	10	4	0.0047	0.0061	1.290	×
501374	bin	34	5	7	0.0045	0.0046	1.028	✓
501420	bin	10	5	0	0.0045	0.0050	1.122	×
501282	cont	16	9	1	0.0042	0.0042	0.997	✓
481263	cont	11	7	1	0.0040	0.0040	0.989	✓
501345	bin	18	5	0	0.0033	0.0040	1.207	×
501206	bin	49	8	4	0.0030	0.0031	1.031	✓
501224	bin	11	9	1	0.0018	0.0071	3.971	×
481836	cont	16	9	11	4.00e-04	4.47e-04	1.116	✓
501408	cont	27	6	0	2.00e-04	3.54e-04	1.772	×
501430	bin	19	10	3	3.17e-09	0.0036	—	×
501298	bin	6	5	0	3.05e-09	—	—	—
501370	bin	10	5	0	2.24e-09	0.0176	—	×
501372	cont	9	5	1	1.78e-09	0.0086	—	×
501326	bin	25	4	0	1.08e-09	0.0028	—	×
480563	bin	8	5	0	9.89e-10	0.0102	—	×
481551	bin	11	6	1	8.77e-10	0.0086	—	×
481942	bin	41	6	7	7.25e-10	0.00e+00	—	✓
474842	bin	45	8	2	4.62e-10	0.0036	—	×
482576	bin	11	5	0	4.54e-10	0.0049	—	×
501407	bin	62	4	0	4.14e-10	0.0041	—	×

ID	Type	k	Treats	MA	$\hat{\tau}_{\text{netmeta}}^2$	$\hat{\tau}_{\text{spring}}^2$	Ratio	Status
501297	bin	52	4	6	3.87e-10	0.0040	—	×
480074	bin	16	5	0	3.77e-10	0.0051	—	×
501434	bin	14	6	3	3.70e-10	0.0039	—	×
480931	bin	13	6	1	3.67e-10	0.0035	—	×
481766	bin	8	7	6	3.02e-10	0.0041	—	×
479996	bin	21	9	2	2.96e-10	0.0047	—	×
501227	bin	21	8	0	1.94e-10	0.0015	—	×
501336	bin	8	4	0	1.85e-10	0.0035	—	×
501312	bin	22	9	3	1.83e-10	—	—	—
501293	bin	19	4	1	1.68e-10	0.0023	—	×
480001	bin	10	7	3	1.59e-10	0.0036	—	×
480052	bin	20	8	2	1.49e-10	0.0024	—	×
482416	bin	8	7	0	1.41e-10	—	—	—
501340	bin	20	4	0	1.04e-10	0.0018	—	×
501207	bin	22	8	1	9.88e-11	0.0021	—	×
479661	bin	14	4	2	9.40e-11	0.0016	—	×
501300	cont	12	4	1	8.80e-11	0.0025	—	×
479807	bin	14	5	0	8.28e-11	0.0019	—	×
501365	bin	13	4	0	8.27e-11	0.0014	—	×
501243	bin	17	7	0	7.97e-11	0.0016	—	×
501197	bin	11	5	2	7.83e-11	0.0015	—	×
501255	bin	29	4	0	7.76e-11	—	—	—
501350	bin	40	11	3	7.68e-11	0.0013	—	×
501309	bin	11	5	2	6.63e-11	0.0013	—	×
482120	bin	16	9	0	5.78e-11	0.0018	—	×
479574	bin	18	7	0	5.16e-11	0.0010	—	×
501325	bin	26	5	0	4.86e-11	9.43e-04	—	×
481571	bin	10	8	1	4.45e-11	0.0014	—	×
482451	cont	11	8	0	3.85e-11	0.0023	—	×
501343	bin	14	5	0	2.98e-11	8.68e-04	—	×
479971	bin	27	4	0	2.35e-11	5.37e-04	—	×
479650	bin	27	6	1	1.87e-11	6.79e-04	—	×
480037	bin	11	7	0	1.53e-11	0.0010	—	×
501252	bin	14	7	0	1.25e-11	8.51e-04	—	×
501419	bin	7	6	0	1.22e-11	0.0022	—	×
482522	bin	6	4	0	9.01e-12	8.46e-04	—	×
482477	cont	5	4	0	6.80e-12	4.08e-04	—	×
476033	bin	7	5	0	6.14e-12	4.84e-04	—	×
501302	bin	8	6	0	6.14e-12	5.17e-04	—	×
481159	bin	8	5	0	5.38e-12	4.93e-04	—	×
501381	bin	12	10	6	4.52e-12	7.48e-04	—	×
501267	bin	26	9	5	3.98e-12	2.36e-04	—	×
482521	bin	6	4	0	3.05e-12	5.68e-04	—	×
501404	bin	23	5	3	2.88e-12	3.19e-04	—	×
480029	bin	17	8	0	2.24e-12	7.68e-04	—	×
501204	bin	27	7	0	1.41e-12	5.20e-04	—	×
482231	bin	7	4	0	3.50e-13	6.90e-04	—	×
479918	bin	15	7	0	2.49e-13	8.39e-04	—	×
482527	bin	10	5	0	2.33e-13	0.0011	—	×